

Frontier Gaps as an Interface Between Clearance Reasoning and Contact-Feasible Multi-Robot Pushing

Abstract—Cluttered path clearing is not only about detecting blockages, but about deciding which width-constrained bottlenecks are worth intervening on and whether they can be opened through executable multi-robot interactions. Existing Navigation Among Movable Obstacles (NAMO) methods mainly reason about obstacle rearrangement at an abstract geometric level, while collaborative pushing methods focus on local contact execution without explicit guidance for multi-obstacle corridor construction. We present frontier gaps as a structured interface that connects these levels by converting reachable clearance bottlenecks into compact, contact-feasible intervention queries. A W -clearance connectivity graph (WCCG) determines whether a traversable corridor already exists and, when it does not, exposes the reachable frontier gaps that block progress. For each selected gap, only gap-conditioned pushing candidates are generated for adjacent movable obstacles, and short-horizon physical simulation selectively validates these induced interventions. Building on this interface, a physics-informed hybrid search prioritizes the most promising candidates. Simulation and hardware results support that frontier-gap-guided planning improves path-clearing reliability while remaining consistent with real multi-robot execution.

I. INTRODUCTION

Large transport vehicles often need to traverse cluttered environments in which direct passage is blocked by movable obstacles such as pallets, boxes, or equipment. In these settings, planning must go beyond collision-free routing: the environment itself must be reconfigured to create a corridor with sufficient clearance [1], [2]. This challenge arises in warehouse automation, disaster response, and other dense workspaces where obstacle interaction is unavoidable. The resulting problem is therefore not only to detect whether passage is blocked, but also to decide how the blockage can be removed through physical interaction.

As shown in Fig. 1, the difficulty becomes sharper when corridor construction is delegated to a team of smaller mobile robots acting on behalf of a larger transport platform [3], [4]. The large vehicle defines the required clearance but does not participate in pushing, so the robot team must identify and execute coordinated contacts that open a traversable corridor for downstream passage. Cooperative pushing is therefore a natural mechanism for path clearing, but it also makes feasibility strongly contact-dependent: pushing one obstacle can induce coupled motion of nearby objects, and whether a candidate action succeeds depends on local geometry, robot size, and obstacle arrangement [5], [6]. As a result, a geometrically plausible clearing sequence may still fail if the required contacts and motions are not executable.

Existing methods typically capture only part of this problem. NAMO-style approaches reason about obstacle rearrangement and corridor creation, but often abstract away the contact mechanics that determine whether an intervention can actually be executed in tightly cluttered scenes [2], [7]. In contrast,



Fig. 1. Frontier-gap-guided corridor opening in simulation and hardware. The robot team selectively opens reachable bottlenecks until a W -clear corridor emerges.

collaborative pushing and multi-robot manipulation methods focus on coordinated contact and force generation, but usually consider single-object transport or more structured settings [8], [9]. Physics-informed planning adds simulation-based validation, yet this validation is often only loosely coupled to the upstream decision of which blockage should be addressed first [10], [11]. For cluttered path clearing, the missing link is therefore a representation that connects scene-level bottleneck selection to executable multi-robot contact planning.

We address this disconnect through *frontier gaps*, which serve as a structured interface between scene-level clearance reasoning and executable multi-robot intervention. They identify where intervention is worthwhile by exposing the reachable bottlenecks that currently block progress, while short-horizon physical validation determines which induced actions are executable. This interface lets cluttered path clearing be treated as a coupled problem of bottleneck selection and physically grounded corridor construction.

II. METHOD

The proposed formulation is organized around a structured interface between scene-level clearance reasoning and contact-feasible multi-robot execution. Given a start-goal query for a large vehicle in a cluttered environment, it first uses a width-aware representation to test whether a W -clear corridor already exists. If passage is blocked, the representation returns a small set of reachable *frontier gaps* that define the current bottlenecks for progress. These bottlenecks are then converted into compact multi-robot intervention queries and selectively validated through physics-informed search. The overall pipeline is shown in Fig. 2.

A. Clearance Representation and Frontier Gaps

We represent the scene with a W -clearance connectivity graph (WCCG), whose role is not to recover a detailed ge-

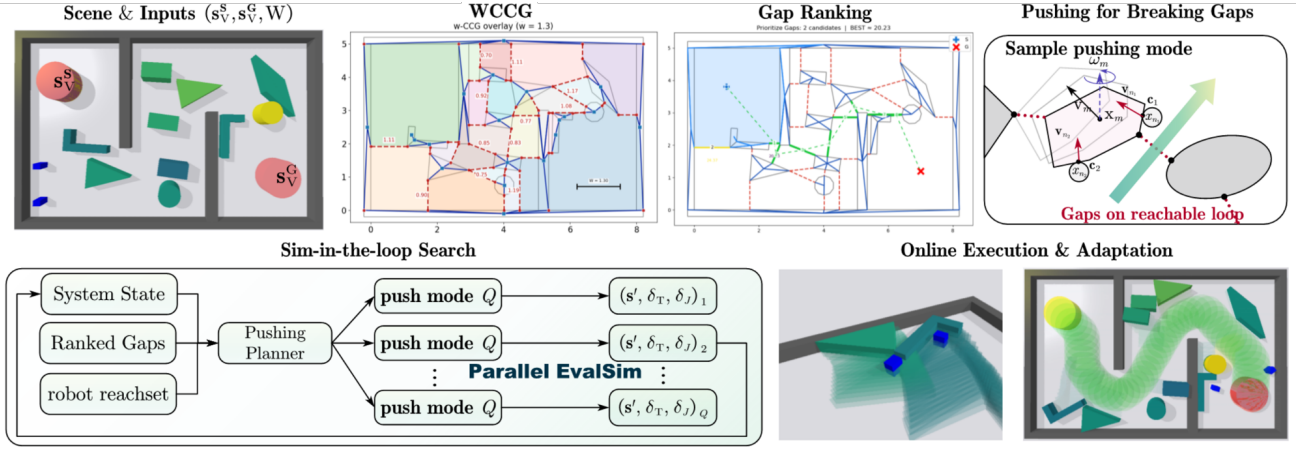


Fig. 2. Overview of the proposed interface-centered formulation. **Top:** The WCCG identifies reachable frontier gaps and induces gap-conditioned pushing candidates. **Bottom-left:** Physics-informed hybrid search. **Bottom-right:** Online execution.

ometric path at every step, but to expose the actionable output needed for path clearing. The graph is constructed directly from obstacle geometry in a width-aware manner. In particular, it uses centroid nodes to represent obstacle components and bridge relations to represent local inter-obstacle passages. The resulting graph is

$$\mathcal{G}_W \triangleq (\mathcal{V}, \mathcal{E}_c \cup \mathcal{E}_b), \quad (1)$$

where $\mathcal{V}$ contains centroid and bridge nodes, $\mathcal{E}_c$ denotes centroid-bridge edges, and $\mathcal{E}_b$ denotes bridge-bridge edges. Each bridge edge is annotated with a local width w_{uv} , defined by the distance between the associated closest points on the two neighboring components. This makes $\mathcal{G}_W$ a compact width-aware representation of where the scene is already traversable and where narrow passages arise.

Given the vehicle start and goal, we perform a width-constrained connectivity query on $\mathcal{G}_W$. If the two lie in the same W -clear connected face, then a feasible corridor already exists and no clearing action is needed. Equivalently, the WCCG answers the following compact connectivity criterion:

$$(\mathbf{x}_V^S, \mathbf{x}_V^G \text{ belong to the same face of } \mathcal{G}_W), \quad (2)$$

If this criterion fails, the query returns the current reachable frontier loop $\mathcal{L}$ that separates the two sides under the width constraint. The bridge-bridge edges visible on $\mathcal{L}$ define the first-hop set of reachable bottlenecks,

$$\Gamma_{\mathcal{L}} \triangleq \{g_1, \dots, g_K\}, \quad (3)$$

where each g_k is a candidate frontier gap exposed by the current clearance query.

These frontier gaps are the key interface produced by the representation. They are not arbitrary narrow passages or generic local heuristics, but the first reachable bottlenecks identified by the width-aware connectivity test. As a result, subsequent stages do not generate contact actions over the full scene. Instead, they operate only on $\Gamma_{\mathcal{L}}$, converting these scene-level clearance bottlenecks into compact contact-feasible intervention queries for multi-robot pushing.

B. Contact-Feasible Pushing Candidates

For a frontier gap $g \in \Gamma_L$ under consideration, the next question is whether this reachable bottleneck can be enlarged through executable multi-robot interaction. The role of this stage is therefore not to solve a fully general manipulation problem, but to convert a scene-level clearance bottleneck into a small set of *contact-feasible intervention queries*.

Let Ω_m denote one of the movable obstacles adjacent to g . For each such obstacle, we sample a small set of short-horizon obstacle motion intents $\{\mathbf{v}_m^k\}_{k=1}^{K_m}$, where $\mathbf{v}_m^k \triangleq (v_x, v_y, \omega) \in \mathbb{R}^3$, biased toward local gap-opening directions while allowing a small number of rotational components to account for local geometry. These sampled directions define the candidate pushing intents for the obstacles adjacent to g .

For each sampled direction $\mathbf{v}_m^k$, we then generate an associated pushing mode $\xi \triangleq (\mathcal{C}_m, \mathbf{u})$, where $\mathcal{C}_m$ denotes the robot-object contact configuration and $\mathbf{u}$ the associated contact wrench parameters. Let $\mathbf{v}_m^{k,B} \in \mathbb{R}^3$ denote $\mathbf{v}_m^k$ expressed in the body frame of Ω_m . The quality of a candidate mode is measured by the primary and multi-directional feasibility losses

$$\begin{aligned} J_F^m(\xi, \mathbf{v}_m^{k,B}) &\triangleq \min_{\mathbf{q}_m, \xi \in Q_{m,\xi}} \|\mathbf{q}_m, \xi + Q_\mu^m(\mathbf{v}_m^{k,B})\|_1, \\ J_{MF}^m(\xi, \mathbf{v}_m^{k,B}) &\triangleq \sum_{\mathbf{v}_d \in \mathcal{D}} w_d \cdot J_F^m(\xi, \mathbf{v}_d), \end{aligned} \quad (4)$$

where $\mathbf{q}_m, \xi \in \mathbb{R}^3$ is the body wrench induced by mode ξ , $Q_{m,\xi} \subseteq \mathbb{R}^3$ is the corresponding admissible wrench set under local contact and friction constraints, $Q_\mu^m(\mathbf{v}_m^{k,B})$ is the quasi-static friction wrench of Ω_m along $\mathbf{v}_m^{k,B}$, and $\mathcal{D}$ with weights w_d augments the desired direction by a small set of neighboring basis velocities. Small J_F^m is necessary for force feasibility, while small J_{MF}^m indicates a more robust mode. Accordingly, the associated mode $\xi_m^k \triangleq (\mathcal{C}_m^k, \mathbf{u}_m^k)$ is obtained by sparse constrained optimization over reachable contacts on $\partial\Omega_m$, minimizing $J_{MF}^m(\xi, \mathbf{v}_m^{k,B})$ under local friction and wrench constraints. Only the best few pairs $(\mathbf{v}_m^k, \xi_m^k)$ are retained, while a persistent *ModeTable* reuses effective entries across similar local geometries. Each retained candidate therefore takes the form $(g, \mathbf{v}, \xi)$, and the gap-induced candidate set is denoted by Ξ_g .

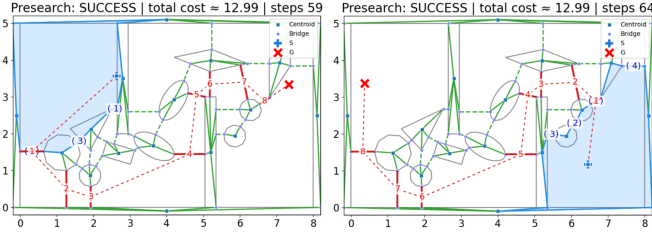


Fig. 3. Frontier-gap ranking under swapped start-goal queries in the same scene. Each panel shows the WCCG with the current reachable face in light blue. Candidate gaps on the frontier loop are ranked by local priority (blue) and refined by short presearch (red), yielding symmetric sequences under the reversed query.

A candidate is retained only if it passes short-horizon physical validation. Specifically, the robots must be able to reach the proposed pre-contact poses, avoid collision during approach and pushing, maintain stable contact, and induce obstacle motion that actually enlarges the target gap. This validation is performed through a short-horizon rollout,

$$(s', \delta_T, \delta_J) \triangleq \text{EvalSim}(s, (g, \mathbf{v}, \xi)), \quad (5)$$

which returns the successor state, time cost, and control effort. As a result, this stage does not enumerate contact actions over the full scene; it converts the reachable frontier gap into a compact set of physically testable multi-robot interventions for the hybrid search.

C. Selective Validation in Hybrid Search

At node ν , the method first checks whether the current state satisfies the connectivity criterion in (2). If not, the updated WCCG induces a reachable frontier loop $\mathcal{L}_\nu$ and the associated candidate gaps $\Gamma_{\mathcal{L}_\nu}$. As illustrated in Fig. 3, these gaps are ranked by the predicted cost-to-connect

$$\widehat{\text{Cost}}(g) \triangleq C(g | \mathcal{L}_\nu, s_R) + C_{\text{pre}}(g), \quad (6)$$

where $C(g | \mathcal{L}_\nu, s_R)$ combines the robots' transition cost to the gap and the estimated effort required to widen it, and $C_{\text{pre}}(g)$ is a short presearch estimate of downstream gap-crossing cost toward the goal. The ranked list is

$$\text{Rank}(\nu) \triangleq \text{argsort}_{g \in \Gamma_{\mathcal{L}_\nu}} \widehat{\text{Cost}}(g), \quad (7)$$

and the node priority is

$$f(\nu) = \chi(\nu) + \min_{g \in \text{Rank}(\nu)} \widehat{\text{Cost}}(g), \quad (8)$$

which balances realized cost and predicted remaining effort. Algorithm 1 summarizes the resulting selective expansion loop.

For each ranked gap $g \in \text{Rank}(\nu)$, the search calls the gap-conditioned candidate generator in Sec. II-B. There, constrained mode generation guided by (4) produces a compact candidate set Ξ_g , and each candidate $\tau \triangleq (g, \mathbf{v}, \xi)$ is evaluated by the short-horizon rollout in (5). PIHS is therefore selective in three senses: the WCCG restricts attention to reachable bottlenecks, gap ranking prioritizes which bottlenecks to test first, and candidate generation together with short-horizon simulation validates only those interventions worth expanding. Repeating this loop yields a corridor-construction schedule that specifies which gaps to open and how to open them.

Algorithm 1: Selective Validation in Physics-Informed Hybrid Search

Input: $s_0, \alpha, \text{EvalSim}(\cdot)$
Output: π^*

- 1 $\nu_0 \leftarrow (s_0, \emptyset), \chi(\nu_0) = 0;$
- 2 Initialize $\mathcal{Q}$ by (8);
- 3 **while** $\mathcal{Q} \neq \emptyset$ **do**
- 4 $\nu = (s, \pi) \leftarrow \mathcal{Q}.\text{pop_min}();$
- 5 **if** s satisfies (2) **then**
- 6 $\pi^* \leftarrow \pi, \text{break};$
- 7 Update the WCCG and construct $\text{Rank}(\nu);$
- 8 **foreach** $g \in \text{Rank}(\nu)$ **do**
- 9 Generate $\Xi_g;$
- 10 **foreach** $(\mathbf{v}, \xi) \in \Xi_g$ **do**
- 11 $\tau \leftarrow (g, \mathbf{v}, \xi);$
- 12 $(s', \delta_T, \delta_J) \leftarrow \text{EvalSim}(s, \tau);$
- 13 **if** success **then**
- 14 $\nu' \leftarrow (s', \pi \cup \{\tau\});$
- 15 $\chi(\nu') \leftarrow \chi(\nu) + \delta_T + \alpha \delta_J;$
- 16 $\mathcal{Q}.\text{push}(\nu');$
- 17 **return** $\pi^*;$

III. RESULTS AND DISCUSSION

We evaluate the proposed formulation in cluttered multi-obstacle clearing tasks with three objectives: to compare it against representative baselines, to assess the contribution of frontier-gap-guided prioritization and structured contact reuse through ablations, and to verify that the validated interventions remain executable on hardware.

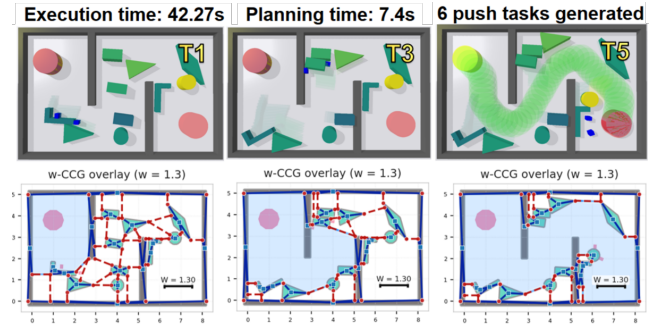


Fig. 4. Representative simulation run with planning-execution alignment. **Top:** PyBullet snapshots of corridor opening by two robots. **Bottom:** corresponding WCCG overlays with the start face shown in blue.

A. Quantitative Evidence

All numerical experiments are implemented in Python3 and PyBullet [12]. As shown in Fig. 4, the nominal setting uses an 8×5 m bounded workspace in which two robots must clear a passage for an external vehicle through scenes containing 10–15 randomly placed movable obstacles. A trial is counted as successful when a W -clear path exists from start to goal and the target reaches its goal disk. All baselines are evaluated under the same W -clearance criterion and contact model.

Table I summarizes the main results. We compare the proposed method against three representative baselines: *DFS-WCCG*, a simulation-in-the-loop depth-first search with fixed

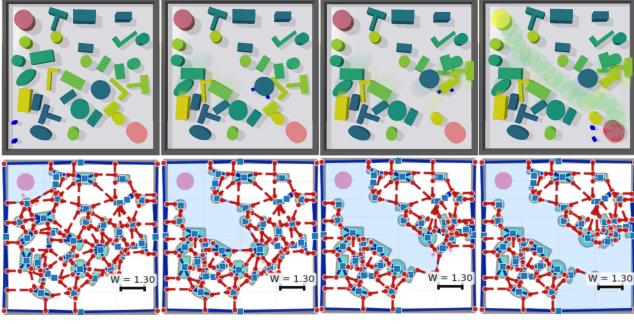


Fig. 5. Scalability in dense clutter with 30 movable objects. **Top**: snapshots during the path clearing of 24.5 s. **Bottom**: WCCG overlays with the active face in blue and ranked frontier gaps.

pushes; *SL-Push*, which clears obstacles along a route and validates the induced pushes with physics; and *Rec-NAMO*, a traditional NAMO-style baseline that improves local traversability incrementally. Across the evaluated scenes, the proposed method achieves the highest success rate at 92.5% while also giving the lowest combined planning and execution time among the successful planners. *DFS-WCCG* suffers from rapid branching growth, *SL-Push* requires longer action sequences and more validation, and *Rec-NAMO* often makes only partial progress without establishing a complete W -clear corridor. Overall, the comparison supports coupling frontier-gap-guided bottleneck reasoning with contact-feasible action generation.

The ablation study clarifies which parts of the formulation drive this performance. Removing `presearch` reduces success from 92.5% to 85.0% while increasing planning time and simulation calls, indicating that frontier-gap-guided prioritization helps avoid unnecessary validation and focus computation on useful bottlenecks. Removing `ModeTable` further reduces success to 80.0%, again with higher planning cost and more simulation calls, suggesting that structured reuse of contact-conditioned action templates improves search efficiency in the same task setting. Together, these ablations support that the full method benefits from both bottleneck-aware prioritization before simulation and structured reuse when converting selected bottlenecks into physically testable action candidates.

Lastly, we examine a denser clutter setting in an 8×5 m workspace with 30 movable obstacles. Despite the increased scene density, the method maintains a compact branching factor: in Fig. 5, it completes the task with 10 node expansions out of 115 visited nodes, 312 short simulation calls, and 7 executed push tasks, yielding a connected W -clear corridor. The planning and execution times are 21.2s and 24.5s, respectively.

TABLE I
PERFORMANCE COMPARISON AND ABLATIONS.

Method / Variant	Succ.(%)	PT (s)	ET (s)	#Sims	#Pushes
DFS-WCCG	25.0	>100.0	51.2	>500	8.0
SL-Push	75.0	18.2	64.6	20.0	10.0
Rec-NAMO	37.5	13.3	42.4	0.0	7.0
FG-Interface (ours)	92.5	10.3	28.6	121.5	6.0
w/o presearch	85.0	16.9	35.2	178	7.0
w/o ModeTable	80.0	15.1	33.4	164	6.0

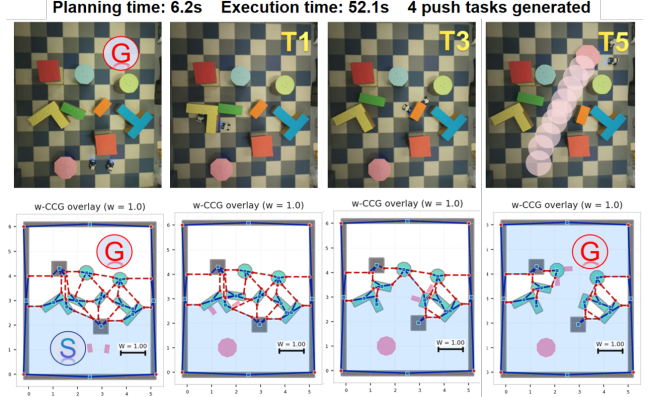


Fig. 6. Representative hardware execution with planning-execution alignment. **Top**: snapshots of two robots sequentially opening frontier gaps in a cluttered workspace. **Bottom**: corresponding WCCG overlays with the current reachable face shown in blue.

B. Hardware Execution

To verify that the frontier-gap-guided interventions remain valid beyond simulation, we deploy the method on a real two-robot platform in cluttered corridor-construction tasks. Each trial is conducted in a 5×6 m workspace with 2 UGVs and 6 movable obstacles, whose observed poses are streamed online for execution and visualization. As shown in Fig. 6, the robots sequentially open frontier gaps until a W -clear corridor emerges from start to goal. Throughout execution, the WCCG is rebuilt from the observed scene configuration, so the planner continues reasoning over the currently reachable bottlenecks rather than following a fixed open-loop obstacle-removal sequence. The representative runs therefore show that the updated WCCG remains aligned with the evolving scene and that the validated gap-opening actions remain executable under real multi-robot interaction.

IV. CONCLUSION AND OUTLOOK

We presented *frontier gaps* as a structured interface for cluttered path clearing, connecting width-aware bottleneck reasoning to executable multi-robot intervention. This formulation enables the planner to focus physical validation on a small set of promising gap-opening actions, supporting reliable corridor construction in both simulation and representative hardware execution.

The current system still depends on sufficiently accurate state estimation and remains sensitive to drift, wheel slippage, and contact variability during real execution. Future work will extend this formulation to settings with limited visibility, uncertain object masses, and tighter coupling between estimation, replanning, and contact execution.

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